Reinforcement Learning

Markov Decision Processes (MDP)

Sutton/Barto* Chapter 3

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With figures from Sutton/Barto*

*Sutton and Barto, Reinforcement Learning: An Introduction, 2nd edition, MIT Press, Cambridge, MA, 2018

Summary of Notation

General

X	capital letters: random variables
x, p	lower-case letters: realizations of random variables or scalar functions
$\boldsymbol{w}$	Bold lower-case letters: real-valued vectors (even if random variables)
$\boldsymbol{W}$	bold capitals: matrices
α	Greek letters: parameters (vectors if in bolt)
$\Pr\{X = x\}$	probability that a random variable X takes on the value x
$X \sim p$	random variable X selected from distribution $p(x) = \Pr\{X = x\}$
$\mathbb{E}[X]$	expectation of a random variable X, i.e., $\mathbb{E}[X] = \sum_x p(x)x$
$\mathrm{argmax}_\mathrm{a}\, f(a)$	a value of action a at which $f(a)$ takes its maximal value

Value Function

G_t	return (cumulative reward) following time t
$G_{t:h}$	return from t to h (discounted and corrected)
$v_\pi(s)$	value of state s under policy π (expected return)
$v_*(s)$	value of state s under the optimal policy
$q_\pi(s, a)$	value of taking action a in state s under policy π
$q_*(s, a)$	value of taking action a in state s under the optimal policy
V, V_t	array estimates of state-value function v_π or v_*
Q, Q_t	array estimates of action-value function q_π or v_*

MDP

s, s'	states
a	an action
r	a reward
$\mathcal{S}$	set of all (nonterminal) states, $\mathcal{S}^+$ are all states
$\mathcal{A}(s)$	set of all actions available in state s
γ	discount-rate parameter
t	discrete time step
T	final time step of an episode (a.k.a. horizon)
A_t	random variable for the action at time t
S_t	random variable for the state at time t
R_t	random variable for the reward at time t
$p(s', r \mid s, a)$	probability of transition to state s' and receiving reward r, from state s taking action a.
$p(s' \mid s, a)$	probability of transition to state s' fom state s taking action a.
$r(s, a)$	expected immediate reward from state s after action a.
$r(s, a, s')$	expected immediate reward from state s to s' with action a.
$\pi(a \mid s)$	probability of taking action a in state s under stochastic policy π
$\pi(s)$	action taken in state s under deterministic policy π

The Agent–Environment Interface

- Markov decision process (MDP) is a mathematical model **for sequential decision making** when outcomes are uncertain.
- The MDP represents a **fully observable, stochastic, and known environment.**
- The goal is specified via rewards.
- Actions influence the future rewards by leading to different states.

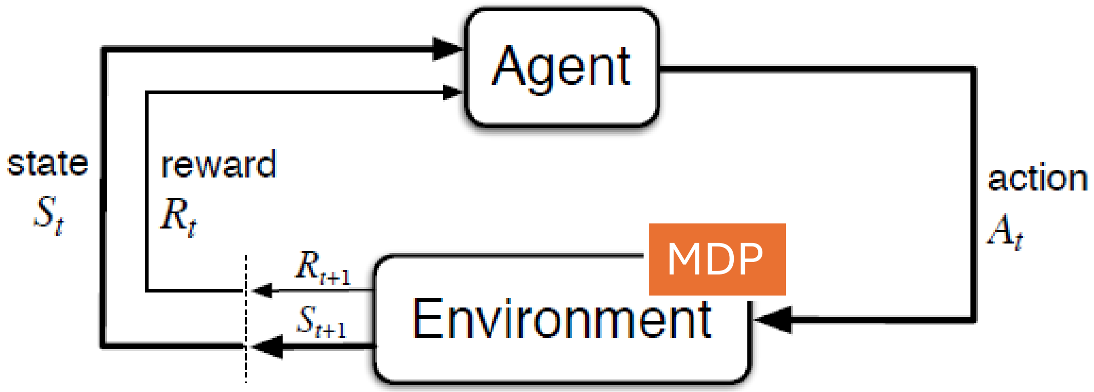


Figure 3.1: The agent–environment interaction in a Markov decision process.

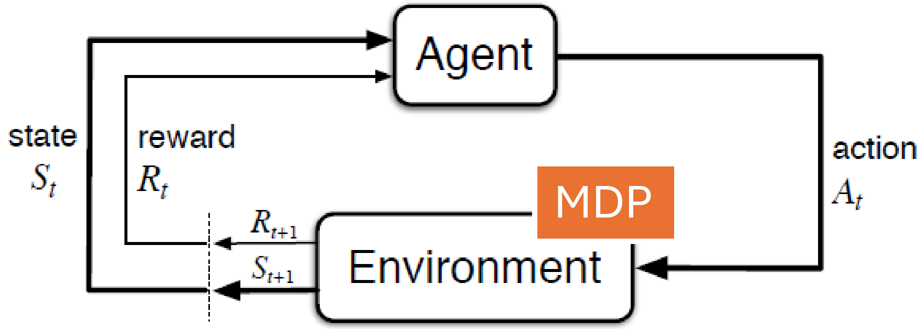
An **interaction** is a sequence of discrete time steps, $t = 0, 1, 2, 3, \ldots$

- At each time step t:
  1. Agent receives some representation of the environment's state $s_t \in \mathcal{S}$
  2. Agent selects an action, $a_t \in \mathcal{A}(s_t)$
  3. One time step later (as a consequence of its action) the agent receives a numerical reward, $r_{t+1} \in \mathcal{R}$ and finds itself in a new state s_{t+1}

Leads to sequence or trajectory that begins like this:

$$S_0, A_0, R_1, S_1, A_1, R_2, S_2, A_2, R_3, \ldots$$

Finite MDP



- General MDPs can have continuous state and action spaces.
- We consider finite MDPs. An MDP is **finite** if the
  - sets of states $\mathcal{S}$,
  - set of action $\mathcal{A}$, and
  - set of rewards $\mathcal{R}$ all have a finite number of elements.

- **Dynamics** of the MDP

$$p(s', r|s, a) \stackrel{\text{def}}{=} \Pr\{S_t = s', R_t = r \mid S_{t-1} = s, A_{t-1} = a\}$$

- Are represented as a deterministic function $p(s, a, r, s')$

$$p : \mathcal{S} \times \mathcal{A} \times \mathcal{R} \times \mathcal{S} \rightarrow [0,1]$$

- A decision process is a MDP if p completely characterizes the environment's dynamics. This implies:
  - Dynamics only depend on the previous state s.
  - The state must have sufficient information about the past. If this is true, then it is called an **information state** and the system has the **Markov property**.

Transition and Reward Functions

- Often the function $p(s', r|s, a)$ is broken down into a transition and a reward model.

- State transition probabilities:

$$p(s'|s,a) \stackrel{\text{def}}{=} \Pr\{S_t = s', |\ S_{t-1} = s, A_{t-1} = a\}$$

$$= \sum_{r \in \mathcal{R}} p(s', r|s, a)$$

- Expected reward for state-action pairs:

$$r(s,a) \stackrel{\text{def}}{=} \mathbb{E}[R_t |\ S_{t-1} = s, A_{t-1} = a]$$

$$= \sum_{r \in \mathcal{R}} r \sum_{s' \in \mathcal{S}} p(s', r|s, a)$$

- Expected rewards for state-action-next-state triples (if the next state also affects the reward):

$$r(s,a,s') \stackrel{\text{def}}{=} \mathbb{E}[R_t |\ S_{t-1} = s, A_{t-1} = a, S_t = s']$$

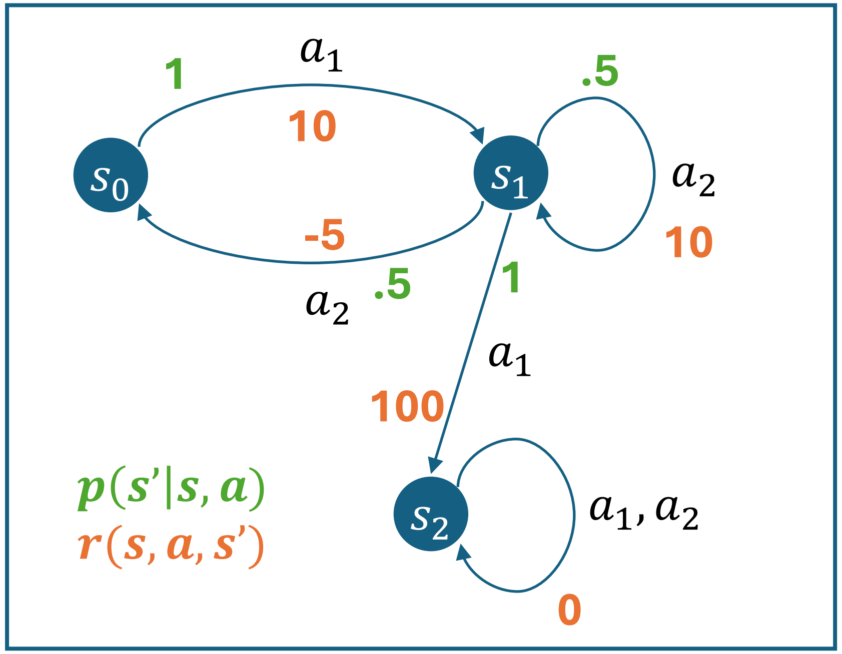
$$= \sum_{r \in \mathcal{R}} r \frac{p(s', r|s, a)}{p(s'|s, a)}$$

Finite Markov Decision Process (MDP)

Finite MDPs are discrete-time stochastic control processes defined by:

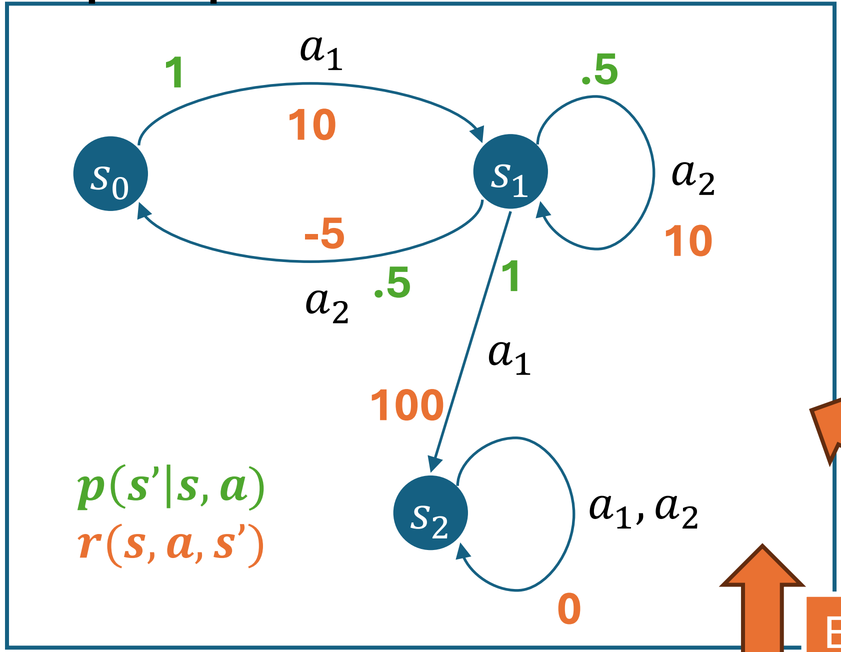
- a finite set of **states** $\mathcal{S} = \{s_0, s_1, s_2, \ldots\}$ (initial state s_0)
- a set of available **actions** $\mathcal{A}(s)$ in each state s
- a **transition model** $p(s' \mid s, a)$ where $a \in \mathcal{A}(s)$
- a **reward function** $r(s)$ where the immediate reward depends on the current state (often $r(s, a, s')$ is used to make modelling easier)

Example MDP as a Graph



Alternative Representations of a Finite MDP

Graph Representation



Equivalent

p-Function Table Representation

s	a	s'	$p(s'\|s,a)$	$r(s,a,s')$
s_0	a_1	s_1	1	10
s_1	a_2	s_0	.5	-5
s_1	a_2	s_1	.5	10
s_1	a_1	s_1	1	100
s_2	a_1	s_2	1	0
s_2	a_2	s_2	1	0

Matrix Representation

Transition matrices

a_1

	s_0	s_1	s_2
s_0	0	1	0
s_1	0	0	1
s_2	0	0	1

a_2

	s_0	s_1	s_2
s_0	1^*	0^*	0^*
s_1	.5	.5	0
s_2	0	0	1

Reward matrices

a_1

	s_0	s_1	s_2
s_0	$-\infty$	10	$-\infty$
s_1	$-\infty$	$-\infty$	100
s_2	$-\infty$	$-\infty$	0

a_2

	s_0	s_1	s_2
s_0	$-\infty$	$-\infty$	$-\infty$
s_1	-5	10	$-\infty$
s_2	$-\infty$	$-\infty$	0

* and $-\infty$ represent illegal actions and transitions!

Example: Mobile Robot MDP

A mobile robot has the job of collecting empty soda cans in an office environment. It has sensors for detecting cans, and an arm and gripper that can pick them up and place them in an onboard bin; it runs on a rechargeable battery. It can actively search for cans or wait for people to drop them off.

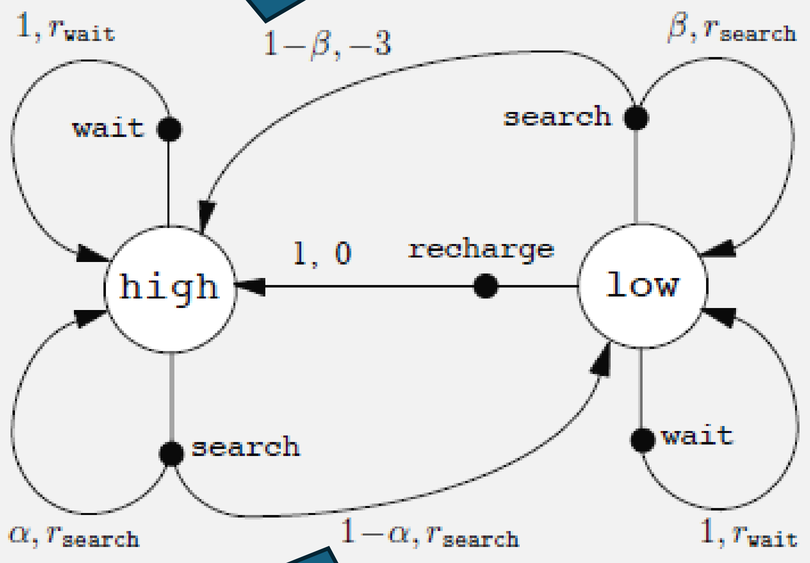
States: $\mathcal{S} = \{\text{high}, \text{low}\}$ // battery charge

Actions: $\mathcal{A} = \{\text{search}, \text{wait}, \text{recharge}\}$

Reward: each collected can gives +1

$1 - \beta$: Chance of running out of battery and needing rescue

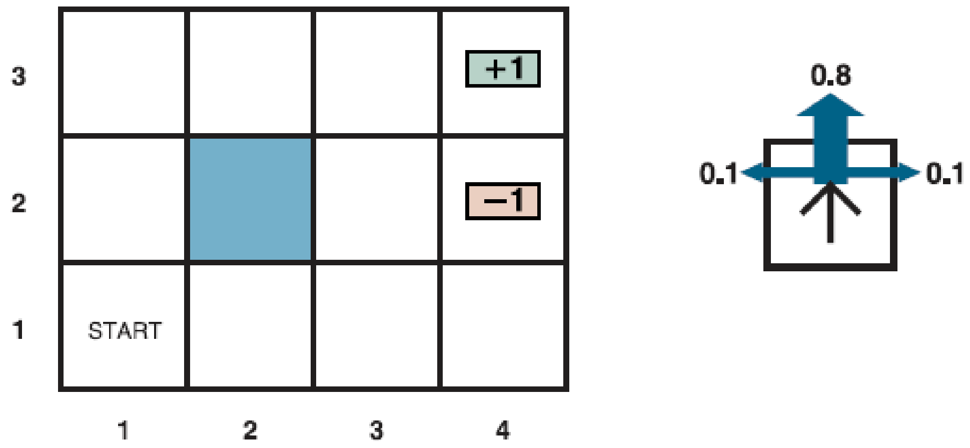
s	a	s'	$p(s' \mid s, a)$	$r(s, a, s')$
high	search	high	α	r_{search}
high	search	low	$1 - \alpha$	r_{search}
low	search	high	$1 - \beta$	-3
low	search	low	β	r_{search}
high	wait	high	1	r_{wait}
high	wait	low	0	-
low	wait	high	0	-
low	wait	low	1	r_{wait}
low	recharge	high	1	0
low	recharge	low	0	-



Expected number of cans collected.

$1 - \alpha$: Battery runs low

Exercise: 3x4 Grid World



(a) (b)

AIMA Figure 17.1: (a) A simple, stochastic environment. (b) Illustration of the transition model of the environment: the “intended” outcome occurs with probability 0.8, but with probability 0.2 the agent moves at right angles to the intended direction. A collision with a wall results in no movement. Transitions into the two terminal states have reward +1 and -1, respectively, and all other transitions have a reward of -0.04.

State representation: (x, y)

1. Draw MDP model as a graph.

2. Complete the following entries in the model as a table

s	a	s'	$p(s'\|s,a)$	$r(s,a,s')$
(1,1)	up	(1,2)		
(1,1)	right	(1,2)		
(1,2)	up	(1,3)		
(3,3)	right	(4,3)		
(3,3)	right	(1,1)		

Goals, Rewards and Returns

Reward Signals

- **Goal**: maximize the expected value of the cumulative sum of the received reward signal.

- The reward signal at time t is a single number:

$$R_t \in \mathcal{R} \subseteq \mathbb{R}$$

- Notes:
  - Reward signals should communicate what the agent wants to achieve and not how to do it. We want to learn how to do it!
  - Typically, only provides a reward for the final goal.
  - Reward shaping can be used to improve learning but may lead to issues (reward hacking).

Return and Episodes

- The agent's goal is to maximize the expected return.
- The return is defined as

$$G_t \stackrel{\text{def}}{=} R_{t+1} + R_{t+2} + R_{t+3} + \cdots R_T$$

We use G so the return is not confused with rewards.

where T is the final step.

- Episodic task:
  - Episodes end with a terminal state at time T.
  - Typically followed by a reset for the next episode.

- Continuing task:
  - Has no episodes! $T = \infty$

Reward Discounting

- Discounting is often used when rewards now are more important then rewards later. Continuing tasks always use discounting.

$$G_t \stackrel{\text{def}}{=} R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

  where $0 \leq \gamma < 1$ is the discount factor.

- Properties:
  - Discounting guarantees a finite sum for G_t if $\{R_k\}$ is bounded for continuing tasks
  - “myopic” agents use $\gamma = 0$ and only maximizes immediate rewards.
  - Larger γ make the agent more farsighted.

- Return is recursive:

$$\begin{aligned} G_t &\stackrel{\text{def}}{=} R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots \\ &= R_{t+1}\, \gamma\, (R_{t+2} + R_{t+3} + \cdots) \\ &= R_{t+1} + \gamma G_{t+1} \end{aligned}$$

Policy and Value Functions

Policy

A (Stochastic) **Policy**: a mapping from states to probabilities of selecting each possible action.

$$\pi(a|s)$$

- Following policy π at time t means to chose action a in state s with probability $\pi(a|s)$

Soft policies

- A stochastic policy is soft if all $\pi(a|s) > 0$.
  I.e., the policy will eventually try all actions.

Stochastic Policy as a Table: $\pi(a|s)$

s	up	right	down	left
(1,1)	0.6	0.1	0.1	0.1
…	…	…	…	…
…	…	…	…	…

Deterministic Policy

Deterministic policy: Prescribes for each state a single action.

$$a = \pi(s)$$

- This just means we have a stochastic policy that has a probability of 1 for exactly one action.

- **Important guarantee**: Finite MDPs always have an optimal deterministic policy.
- **Intuitive reason**: If there is a best action given all we know (i.e., we know the current state), then there is no point in choosing a suboptimal action.

Deterministic Policy as a Table

s	Action $\pi(s)$
(1,1)	up
…	…
…	…

equivalent

Stochastic Policy as a Table: $\pi(a|s)$

	up	right	down	left
(1,1)	1	0	0	0
…	…	…	…	…
…	…	…	…	…

Value Function

- A policy $\boldsymbol{\pi}$ defines for each state which action to take.
- **State value function**: Expected value of being in a state when following π.

$$v_\pi(s) \stackrel{\text{def}}{=} \mathbb{E}_\pi[G_t \mid S_t = s]$$

$$= \mathbb{E}_\pi\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \middle| S_t = s\right] \text{ for all } s \in \mathcal{S}$$

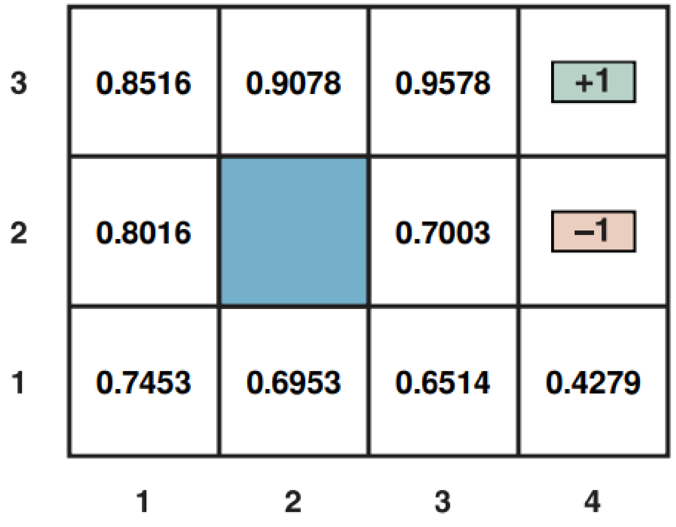
- Representation as a tabular estimate V:

γ .. Discounting factor to give more weight to immediate rewards.

$\mathbb{E}_\pi$ ... Expectation over sequences that can be created by following π.

R_t .. Reward at time t

Value Function for the 3x4 Grid World



Value Function

s	**State Value $V(s)$**
(1,1)	0.7453
(1,2)	0.8016
…	…

Action-value Function: Q-function

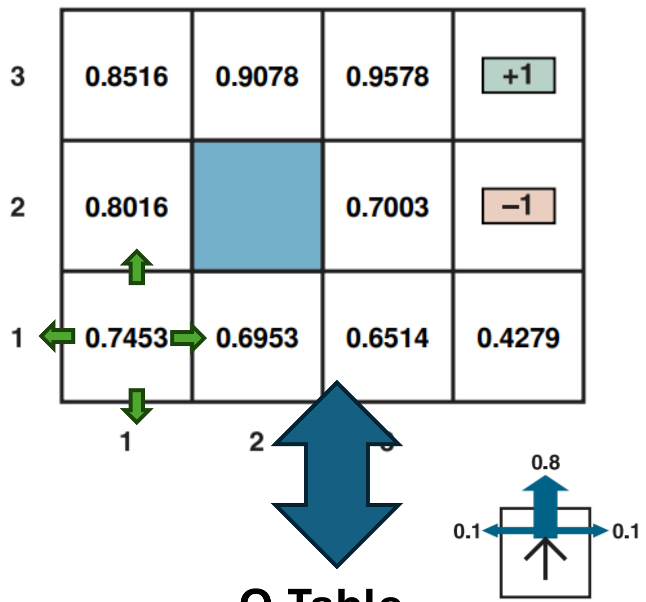
- Expected value of choosing an action in a state and then following π.

$$q_\pi(s,a) \stackrel{\text{def}}{=} \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a]$$

$$= \mathbb{E}_\pi\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \middle| S_t = s, A_t = a\right] \text{ for all } s \in \mathcal{S}$$

- The Q-function lets us compare the value of taking different action is a given state. It is used in algorithms to determine what action is the best (the action with the highest Q-value for the state).

Value Function



Q-Table

s	a	$q(s,a)$
(1,1)	up	0.7453
(1,1)	right	0.6709
(1,1)	down	0.7003
(1,1)	left	0.7109
...	...	...

Value Function Estimation using Monte Carlo Methods

- The value functions v_π and q_π can be estimated from experience. This is called **policy evaluation**.

- **Sample-average method** (a simple Monte Carlo method):
  - Follow policy π.
  - Maintains an average $\bar{V}(s)$ of the actual returns that have followed that state. The average is updated at each visit.
  - Convergence: $\lim_{visits \to \infty} \bar{V}(s) = v_\pi(s)$

- This also works for $q_\pi(\mathrm{s}, \mathrm{a})$

- For problems with many states: Instead of storing all $\bar{V}(s)$ in a table, learn an **approximate function** with a small number of parameters.

Calculating the Value Function

Value functions always have a **recursive relationship:**

Value of a state = immediate reward + discounted value of successor state

The **Bellman (Expectation) Equations** for a given policy:

There are $|\mathcal{S}|$ Bellman equations.

$$
\begin{aligned}
v_\pi(s) &\overset{\text{def}}{=} \mathbb{E}_\pi[G_t | S_t = s] \\
&= \mathbb{E}_\pi[R_{t+1} + \gamma G_{t+1} | S_t = s] \\
&= \underbrace{\sum_a \pi(a|s) \sum_{s',r} p(s',r|s,a)}_{\text{Expectation}} [\underbrace{r}_{\text{Reward}} + \underbrace{\gamma v_\pi(s')}_{\text{Value of next state}}] \quad \text{for all } s \in \mathcal{S}
\end{aligned}
$$

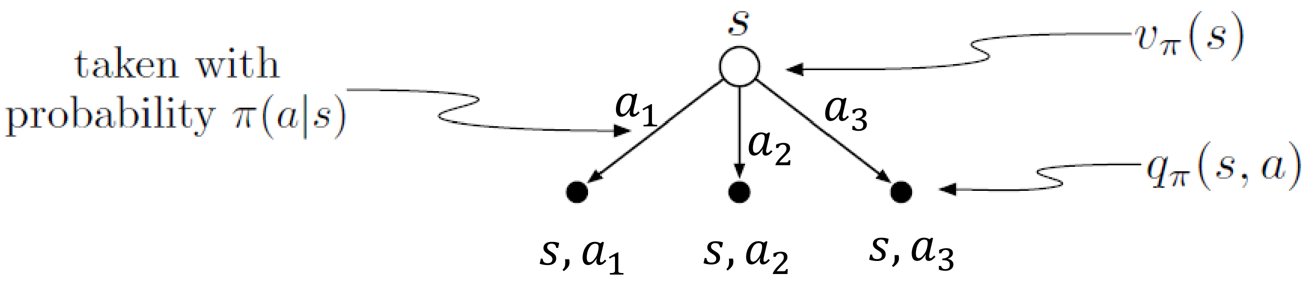
Alternative formulation with transition and reward function:

$$v_\pi(s) = \sum_a \pi(a|s) \sum_{s'} p(s'|s,a)[r(s,a,s') + \gamma v_\pi(s')] \quad \text{for all } s \in \mathcal{S}$$

Backup Diagrams: Notation

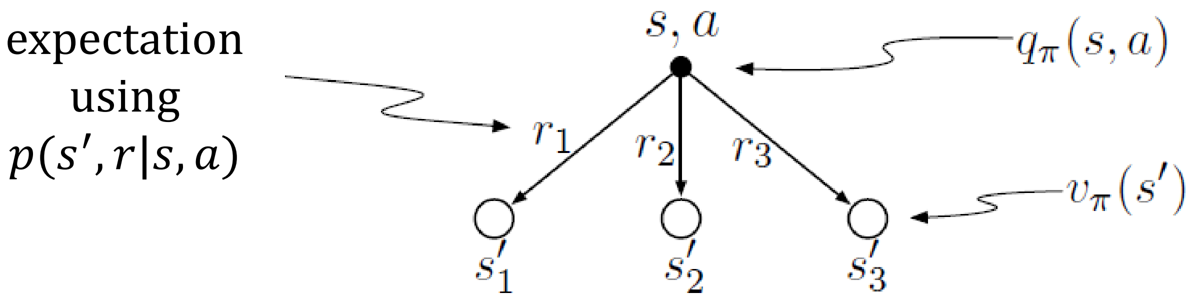
- The relationship between state values, action probabilities, state-action values and expected rewards can be visualized as a tree.

Choosing an action in a state (= calculate the expectation over actions)



$$v_\pi(s) = \sum_a \pi(a|s)\, q_\pi(s, a)$$

Calculate the expected state-action value



$$q_\pi(s, a) = \sum_{s', r} p(s', r|s, a)[r + \gamma v_\pi(s')]$$

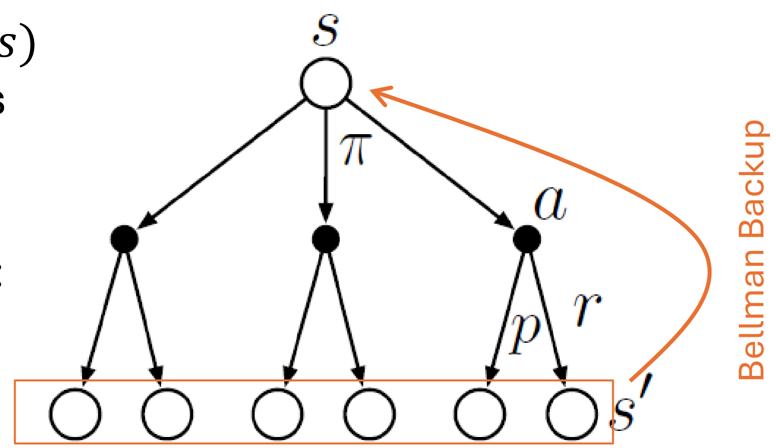
Bellman Equation as a Backup Diagram

$$v_\pi(s) \overset{\text{def}}{=} \overbrace{\underbrace{\sum_a \pi(a|s)}_{\text{Choose action}} \underbrace{\sum_{s',r} p(s',r|s,a)}_{\text{Expectation for transition and reward}}}^{\text{Expectation}} [\underset{\text{Reward}}{r} + \gamma \underbrace{v_\pi(s')}_{\text{Value of next state}}] \text{ for all } s \in \mathcal{S}$$

Backup diagram for $v_\pi(s)$ shows all possible paths for the expectation.

All paths are of the form:

s, a, r, s'



If $v_\pi(s)$ is estimated using $v_\pi(s')$ estimates, then this is called **bootstrapping**.

Code...

Optimal Policies and Value Functions

The Optimal Policy

- Value functions define a partial ordering over policies.
  A policy is better if it has at least one larger state value.

$$\pi' \geq \pi \Leftrightarrow v_{\pi'}(s) \geq v_\pi(s) \text{ for all } s \in \mathcal{S}$$

- There always exists an **optimal policy** π_* with the optimal state-value function v_* defined as

$$v_*(s) \stackrel{\text{def}}{=} \max_\pi v_\pi(s) \text{ for all } s \in \mathcal{S}$$

- This also implies the existence an optimal action-value function:

$$q_*(s,a) \stackrel{\text{def}}{=} \max_\pi q_\pi(s,a) \text{ for all } s \in \mathcal{S}, a \in \mathcal{A}$$

  Since q_* directly depends on the optimal value function v_*

$$q_*(s,a) = \mathbb{E}_\pi[R_{t+1} + \gamma \boxed{v_*(S_{t+1})} | S_t = s, A_t = a]$$

- If we know all $q_*(s,a)$ values, then we can determine the best action for each state by the highest Q-value. It is optimal to always choose this best action. **MDPs always have an optimal deterministic policy!**

Bellman Optimality Equations

- **Idea**: π^* always picks the best action, so the optimal state value needs to be the state-action value of the best action.

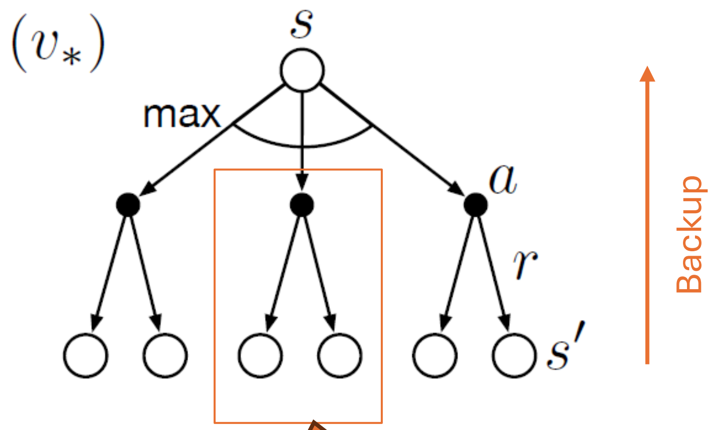
$$v_*(s) = \max_{a \in \mathcal{A}(s)} q_*(s,a) \qquad \text{for all} \quad s \in \mathcal{S}$$

- The Bellman optimality equations can be rewritten as:

$$\begin{aligned} v_*(s) &= \max_{a \in \mathcal{A}(s)} q_*(s,a) \\ &= \max_a \mathbb{E}_\pi \left[G_t | S_t = s, A_t = a\right] \\ &= \max_a \mathbb{E}_\pi \left[R_{t+1} + \gamma G_{t+1} | S_t = s, A_t = a\right] \\ &= \max_a \mathbb{E}_\pi \left[R_{t+1} + \gamma v_*(S_{t+1}) | S_t = s, A_t = a\right] \\ &= \max_a \sum_{s',r} p(s',r\,|s,a)[r + \gamma v_*(s')] \end{aligned}$$

Issue with recursive relationship: To calculate $v_*(s)$, we need to know $v_*(s')$

Backup diagram with max operator



Pick the expected value for the best action

Bellman Optimality

Max is a non-linear operator.

- The optimality condition is a system of $|\mathcal{S}|$ non-linear equations of the from:

$$v_*(s) = \max_{a \in \mathcal{A}(s)} q_{\pi_*}(s, a) \quad \text{for all } s \in \mathcal{S}$$

- The problem can be linearized as a LP using constraints.

- **Issues** with solving the system:
  - The **Markov property** for states has to be satisfied.
  - The **transition and reward model** (function p) needs to be completely known.
  - Computational resources for **solving a large system of non-linear equations.**

- This is rarely possible in real applications and we need approximate methods.

Summary: Relationships between Value Functions and Policies

Value functions are defined as expected returns following a policy:

$$v_\pi(s) \stackrel{\text{def}}{=} \mathbb{E}_\pi[G_t | S_t = s]$$

$$q(s, a) \stackrel{\text{def}}{=} \mathbb{E}_\pi[G_t | S_t = s, A_t = a]$$

- For the optimal policy and value function we have:

$$q_*(s, a) \stackrel{\text{def}}{=} \mathbb{E}_{\pi_*}[G_t | S_t = s, A_t = a]$$

$$\pi^*(s) = \operatorname*{argmax}_{a \in \mathcal{A}} q_*(s, a)$$

Related Models

MDP simplifications:

- **Multi-armed bandits (Chapter 2 in Sutton/Barto)**
  - You repeatedly choose an arm (action).
  - You immediately get a reward and start over.
  - There is no state evolution.

- **Markov Reward Process**: A Markov chain with rewards (no actions).

MDP extension:

- Partially Observable Decision Process (POMDP): a MDP for partially observable problems. It adds:
  - Observations O
  - Observation probabilities $Z(o|s, a)$

Many other related models were developed: Semi-Markov decision processes, contextual bandits, Markov games, constrained MDPs, multi-objective MDPs, Dynamic Bayesian Networks

Summary: What You Should Know

Terms

- **Actions** are the agent's choices
- **States** are the basis for the choices
- **Reward** is used to evaluate choices
- **Objective**: maximize expected return
- **Policy**: a stochastic rule to select actions as a function of the state
- **Information state** = **Markov property** for states

Backup Diagrams to represent the calculation of expected values.

Bellman optimality Equations

- **Recursive relationship** between state values.
- MDPs always have an **optimal deterministic policy**.

Tasks we will talk in the next sessions:

- **Policy evaluation** (prediction): estimate v_π or q_π for a given fixed π.
- **Control**: find optimal π_*